

UTrucking AI Phone Assistant — Plan, Progress & Roadmap

Last updated: 2026-07-09 · **Live agent:** v43 · **Backend:** latest pushed (Round 16 on `main` ; one-click Render deploy activates the newest tools)

How to read this. Four parts, in order: **1 — What's been done** (built, tested, live). **2 — Where the value is added** (in dollars, from our own data). **3 — What's next** (the roadmap). **4 — What's required** to get there (a phone number, SMS, accounts, and their costs).

Part 1 — What's been done

The assistant is **live and fully tested**. It answers calls in a warm, natural voice, looks up any student's order across **both** data sheets, verifies the caller's identity before sharing anything, answers one question at a time, quotes prices on the call, handles general questions from a knowledge base, and transfers to the team on request. The backend is fixed and current: the item/invoice sheet now loads (**654 records**) and the name-match was tightened to reduce wrong-person matches.

Delivered & live

Capability	Status
Order lookup by name (handles mispronounced / misspelled names)	✓ Live
Natural, one-answer-at-a-time conversation (no rambling)	✓ Live
Identity verification — name plus a second detail	✓ Live
Privacy protection — never reveals another student's data	✓ Live
Two-sheet join (dispatch logistics + invoice/item detail)	✓ Live
Instant price quote on the call — <i>"five boxes and a mini fridge" → "about \$133"</i>	✓ Live (v34)

Capability	Status
Busy-day steering — offers open days when a caller asks for a slammed one	✓ Live (v34)
Pricing / service questions from the knowledge base	✓ Live
Transfer to the office at (314) 266-8878	✓ Live
Safety guardrails (stays on-task, resists misuse)	✓ Live
Post-call logging (found, order ID, resolution, sentiment)	✓ Live
Noise-proof listening — cancels background speech in a busy dorm hallway	✓ Live (v43)
Says the dorm names correctly — a pronunciation dictionary for Umrath, Liggett, Nemerov, Lopata...	✓ Live (v43)
Keypad entry — a caller can type their order # or phone last-4 instead of speaking it	✓ Live (v43)
Voicemail detection — leaves a short callback message instead of talking to a machine	✓ Live (v43)
Backup voice — the line keeps working if the speech provider has an outage	✓ Live (v43)
Stays on topic — politely declines off-topic questions instead of guessing	✓ Live (v43)

Built & ready (activate on the next one-click deploy)

Capability	Status
One front-facing dashboard — all tools behind six cards (chat, voice, estimate, ask-your-data, insights, ops); Back button / Esc to return. Redesigned: animated starfield, orbit hub, glass cards, one brand family across every tool page, fully mobile-checked	● Built — deploy to go live
Estimate: photo + description together — typed counts override the photo, extra typed items are added, every line shows its source; typo-tolerant parser with visible closest-match ("you said X")	● Built — deploy to go live
Any-item pricing (AI matching) — items not on the price list are matched to the closest priced item and shown transparently; verified on an 80-item student-goods	● Built — deploy to go live

Capability	Status
gauntlet (80/80, nothing dropped). AI calls ride a 3-model fallback chain so free-tier rate limits don't take tools down	
Customer self-serve estimate page — snap a photo <i>or</i> type items → instant price	● Built — deploy to go live
Photo-to-quote (AI vision) — detects items in a photo and prices them (free Gemini tier)	● Built — deploy to go live
Web chat assistant (SMS preview) — quotes, pickup dates & identity-verified order lookup; no phone number needed	● Built — testable now
Browser voice assistant — same brain, spoken in the browser (free, no Retell minutes)	● Built — deploy to go live
Business insights dashboard — live revenue, top items, upsell pairs, demand, funnel, data-quality scorecard	● Built — deploy to go live
Ask-your-data staff copilot — plain-English questions on aggregate stats (refuses individual-customer data)	● Built — deploy to go live
Billing-leakage audit — flags \$0 / missing-invoice / missing-order rows	● Built
Dispatch / route planner — clusters a day's pickups by building and splits crews	● Built
Ops Command Center — staff page: pick a day → balanced crew routes + printable run sheets (staff-key-gated)	● Built — deploy to go live
Next-season demand forecast — projects the peak window, crews needed, and the fall return season from this year's shape; shown as an Insights planner card	● Built — deploy to go live
Repeat-customer multi-order lookup — a caller with several orders picks which one (by order #, service, or month) before the identity gate	● Built — deploy to go live
Upsell on every quote — every quote suggests the add-ons students actually store together ("most people also add a Plastic Container or Mini Fridge"), never an item already in the basket	● Built — deploy to go live

Capability	Status
Caller-ID lookup — the order lookup can resolve a returning caller by their on-file phone number (identity gate still applies); auto-greeting on inbound waits on a provisioned phone line	● Built — capability ready
Damage / condition photo check (/condition) — a photo returns a good/wear/damage read with notes: move-out dispute protection + a protection-plan upsell	● Built — deploy to go live
Staff console (/staff) — one morning-standup page: today's run sheet, revenue-to-recover flags, demand forecast, data health	● Built — deploy to go live
Sheet caching + resilience — a short cache in front of both sheets serves the last good copy through a transient Google outage, so a live quote/lookup never blanks	● Built — live-safe
Automated test suite + CI — 172 offline tests (engines, endpoints, adversarial edges) run on every code push via GitHub Actions	● Built
Voice QA — every call auto-scored — the moment a call ends, an AI reviewer grades it (identity gate held? over-promised? wrong info? caller frustrated?) and staff see a scoreboard with score, sentiment, response speed and cost per call	● Built — deploy to go live
Release gate — 12 simulated callers — impersonators, prompt injections, third-party callers and every verification path are replayed against each new version of the assistant before it can go live	● Built
Your business data inside Claude — the backend doubles as a secure connector so the owner can ask questions in plain English from their phone (aggregate data only; no customer details)	● Built — deploy to go live

Privacy note (identity gate). Order lookups verify a **second detail** — the caller's building or the last-4 of their phone — before any personal data is shared, on both the phone line and the web chat. This works today with **no phone number**. **Future option:** once a texting number is live, this gate can be upgraded to a **one-time SMS code** (the strongest identity check) with no change to the rest of the flow.

Documentation

Flow maps, connections inventory, QA log, executive deck, and a full **Data & Revenue Audit** — all in this report.

Part 2 — Where the value is added

Every number below is from UTrucking's own dispatch and invoice data (see the **Data & Revenue Audit**).

- **It protects the money-days.** \$87,782 was invoiced in a **13-day** move-out sprint, and **74% of it landed in just 5 days**. When a caller asks for one of those packed days, the assistant now steers them to an open day — so the peak stops overflowing and dropping orders.
- **It quotes instantly, consistently.** The box drives **65% of revenue** on **96.9%** of orders. The assistant (and the new estimate page) price it the same way every time — no under-quoting, no waiting for a callback.
- **It turns your photo data into a moat.** UTrucking already collects item photos at pickup. The **photo-to-quote** tool turns that into an instant customer-facing estimate no competitor can easily copy.
- **It plugs the leaks.** The audit found **~\$1,056** of billing leakage (\$0 / missing-invoice orders) and **212** un-dispatched orders. The billing audit surfaces the leakage today; the booking flow (Part 3) stops it at the source.
- **It deflects the repetitive calls.** "Where's my stuff / what did I order / how much" now get answered without a person — the front desk stops repeating itself.

The single biggest lever, from the data: raise the box price **\$2 → +\$5,186/year (+5.9%)** at near-zero risk. It's a management decision; the assistant then quotes the new price automatically.

Part 3 — What's next

The theme: turn the assistant from **answering** into **doing**. We build in three outcome-focused waves, one tool at a time, each tested against real data before the next. **Wave D is logged for later.**

Wave A · Win more bookings ● **building first**

Tool	What it does
Customer estimate page (photo or text)	Self-serve instant quote — <i>built, deploys next</i>
Web + SMS booking assistant	The phone agent's brain, on text and web chat — students text, they don't call
Spanish parent line	Multilingual voice so parents can handle logistics & payment
Group / referral booking	Roommates / floors book together — dorms already cluster, cutting cost per stop

Wave B · Survive the peak ● after A

Tool	What it does
Book / reschedule / cancel on the call	The receptionist replacement — the biggest staff-time win
Reminders & confirmations	"Your pickup is tomorrow — reply to reschedule" — cuts no-shows
Auto-dispatch + routing	Clusters orders by building, sequences stops, assigns crews — attacks the 212-order backlog
Movers' field app	Crew route + item lists + photo/complete, auto-writes status back — ends "unknown building"

Wave C · Stop leaking money ● after B

Tool	What it does
Invoice automation + leakage guard	Auto-invoice; block \$0 / missing-invoice orders — recovers the ~\$1,056 gap
Payment chaser	SMS follow-up with a pay link — faster cash
Damage / condition vision docs	Auto-tags item condition from photos already taken — dispute protection + a protection-plan upsell

Wave D · See & predict ● built

The **live ops dashboard** (Business insights), the **"ask-your-data" staff copilot**, the **Ops Command Center** (daily crew routing + run sheets), and a **next-season demand forecast** are

all **built and verified against the live sheets**. The forecast projects the peak window and the crews it needs (live: the peak day runs **334 pickups** → **~23 crews' worth of work**), the move-out-window share, and the fall **return season** (Rental Returns ≈ 13–25% of volume). Repeat customers with several orders are now disambiguated before the identity gate. Still logged for later: fall return-season **outbound automation** ("want your stuff back?" SMS), which needs booking write-back + a texting number.

The four moves the AI caller itself will run

All four unlock from the **same booking capability** (Wave B) — build it once, the assistant executes all of them:

1. **Smooth the peak** — offer the nearest open day first, spreading load off the two days that hold 74% of revenue.
2. **Upsell the basket** —  *built*: every quote already offers the add-ons the data shows people take (Plastic Container \$18, Mini Fridge \$23). Booking (Wave B) lets the assistant actually add them to the order.
3. **Close the leak** — never finalize a storage booking with a \$0 total or missing invoice.
4. **Clean the data at the source** — confirm the correct building and a working phone at booking, killing the "unknown building" rows and enabling caller-ID next season.

(Two audit moves are not phone tasks: setting the box price is management's call; clearing the dispatch backlog is the ops team's.)

Part 4 — What's required to go further

What's built runs on the current setup at **no extra cost**. Everything below needs a simple account — costs are small and mostly usage-based.

To enable...	You provide	Rough cost
Quote engine · estimate page · scheduler · route planner · billing audit	Already built on the current backend	\$0 extra
A real phone line for the assistant	A phone number in Retell (or port a Twilio number)	~\$1–3 / mo + ~\$0.07–0.10 / min

To enable...	You provide	Rough cost
Text / SMS (reminders, confirmations, pay-links)	A Twilio account + number	~\$1–2 / mo + ~\$0.008 / text
Photo-to-quote & damage-photo tagging (AI vision)	A vision-AI key — free tier (Google Gemini)	\$0 on the free tier
Booking / rescheduling + auto-invoices	A Google service account (edit access) <i>or</i> a small database	free
Card payments / pay-links	A Stripe account	per-transaction fee

The key unlock is booking write-back. Reading the sheets is free and live; *writing* to them (to book, reschedule, or invoice) needs the free Google service account or a small database. That one piece turns on most of Wave B and C.

To go live on a real phone line: provision a Retell number (budget a one-time ~\$20 for the number + first-month testing), then the caller-ID greeting and outbound reminders become possible.

Appendix — Future-proofing before scale

Small investments now that prevent expensive breakage later:

Area	Today	Risk if left	Recommended before scale
Deploys	Render auto-deploy is OFF ; deploys are manual	A fix silently never ships	Re-enable auto-deploy or wire a deploy hook
Data store	Google Sheets read as CSV per call	Fine for reads; can't handle concurrent writes (needed for booking)	Move to Postgres on Render (free/\$7 tier) when booking lands
Uptime	Render free tier spins down → cold start on first call	First caller of the hour waits ~30–60s	<code>keep_alive</code> helps; a paid instance (~\$7/mo) removes cold starts
Backend access	Endpoints are open (<code>CORS *</code> , no auth)	Anyone can call the API	Add a shared secret/header check

Area	Today	Risk if left	Recommended before scale
Join key	Name-based join across two sheets	Multi-order customers can mix records	Join on <code>order_id</code>
Backups	Live sheets only	Accidental edit = data loss	Nightly export/backup of both sheets
Secrets	Keys in Render env vars	Exposure if a repo is public	Keep in env vars; rotate periodically; keep the backend repo private